Innovative financing mechanisms that enable governments to share risks with private entities can also go a long way in mobilising private sector finance. And expanding public climate finance, particularly in low-income countries, can deliver significant returns at relatively low costs.



STRENGTHEN FOOD SECURITY

The IPCC estimates that in the next decade alone, climate change will drive 132 million more people into extreme poverty. Global warming will jeopardise food security, exacerbate malnutrition, and increase the incidence of heat-related mortality, heart disease and mental health challenges. Species and ecosystems will face dramatic changes as well, such as mangroves failing to counteract rise in sea level, decline in sea ice-dependent species and large-scale tree deaths.

Conservation of flora and fauna is imperative. An ecosystem-based adaptation is inevitable. This approach encompasses a wide range of strategies, from the protection, restoration and sustainable management of ecosystems to more sustainable agricultural practices such as integrating trees into farms, increasing crop diversity and planting trees in pastures. Ecosystem-based adaptation minimises climate risks, including droughts, extreme heat, floods and fires, while also delivering co-benefits for biodiversity, livelihoods, health, food security and carbon sequestration.



RECONFIGURE SOCIAL PROGRAMMES

The report breaks new ground on the feasibility, effectiveness and potential of climate adaptation measures to deliver co-benefits like improved health outcomes or poverty reduction. Cash transfers, public works programmes and social safety nets to include adaptation can lower the vulnerability of urban and rural communities to a wide range of climate risks. Equally vital is improving access to infrastructure and basic services, such as clean water, sanitation and healthcare. Providing these services improves the climate resilience of vulnerable communities.

Emerging evidence suggests that coupling nature-based solutions with engineered options like flood control channels may help reduce water-related and coastal risks. Access to better technologies, such as more resilient crop varieties, improved livestock breeding, or solar and wind power, can also help strengthen climate resilience.



PROMOTE CARBON CREDITS

The carbon credit market is poised for accelerated growth, especially with the commitment made at the 2021 United Nations Climate Change Conference (COP26) in Glasgow to achieve 'net zero' emissions by 2070. The carbon credit trade could help generate funds for green energy projects. Between 2020 and 2070, India alone would need an estimated $10.1 trillion investment in the power, hydrogen and mobility sectors. It is now the largest exporter of carbon credits.

One challenge, however, is that carbon pricing remains a mere footnote in international climate negotiations. Currently, there is no single market for carbon credits. Besides sovereign markets, there is also the voluntary market. Private companies prefer the latter, where prices vary from as low as 10 cents to $80. Article 6 of the Paris Agreement provides a framework for international cooperation on emission reductions. One of COP26's major achievements was the finalisation of the Paris Rulebook—a set of decisions detailing the implementation of the agreement. This is expected to unlock the potential of international emissions trading and contribute to the exponentially growing market of voluntary carbon credits, whose value topped $1 billion in 2021.

McKinsey estimates that demand for carbon credits could rise by a factor of 15 or more by 2030 and by a factor of up to 100 by 2050. Given the demand for carbon credits that could ensue from global efforts to reduce GHG emissions, the world will need a large, transparent, verifiable and environmentally robust voluntary carbon market. The market today is fragmented and complex.

India has the potential to be one of the largest carbon credit markets globally, as the government pushes for a uniform carbon trading market. The appetite for carbon credits will evolve over the next couple of years, as global demand and pressure for 'net zero' industries grow. The growth of carbon credit trade depends on the formation of requisite market systems and a National Carbon Registry to meet a country's commitments under the Nationally Determined Contribution.

Clearly, the COP27 summit to be held in Egypt in November is a crucial opportunity for governments to make progress on all fronts, and for developed countries to demonstrate their solidarity with vulnerable nations where 3.6 billion people live. Everyone involved must step up. It is the only way forward. ■

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